# Results

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## 5 Results

### 5.1 Macro Trends

Figure 1 shows the monthly share of congressional deficit tweets, by party, from June 2017 to January 2023. Consistent with best practices in political science, this paper will represent the Republican Party with red shading and the Democratic Party with blue. Shaded areas indicate legislative periods – defined as the month before, during, and after the passage of a fiscally significant bill. Legislative periods exclude small appropriations, procedural votes, and commemorations.

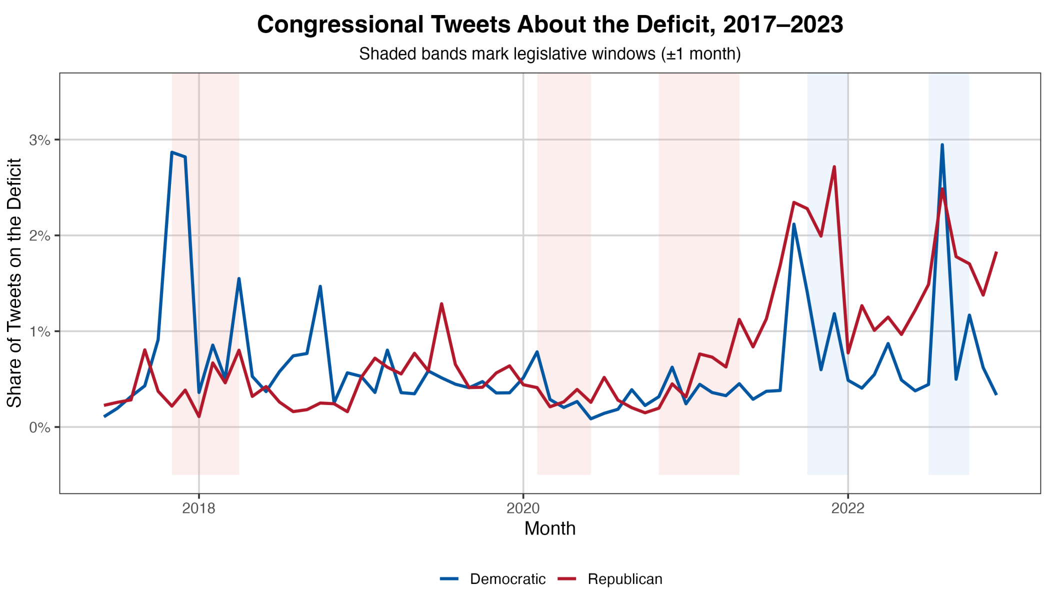


Figure 1: Share of congressional tweets about the deficit.

The baseline deficit tweeting share for both parties is low, generally under 1% when averaged across members over six years, with notable accelerations occurring around legislative periods. Democratic tweeting patterns remain relatively consistent across the six years, with three notable peaks, all during legislative periods: December 2017 (Tax Cuts and Jobs Act), October 2021 (Infrastructure Investment and Jobs Act), and August 2022 (Inflation Reduction Act, CHIPS & Science Act). Republicans, however, demonstrate consistently higher deficit tweeting rates as the minority party, as illustrated in the peaks (2022, 2023) compared to relatively low rates from 2017-2021.

## 5.2 Economic Indicators

Figure 2 plots deficit tweet share against US CPI inflation (YoY) reveals a relatively strong positive relationship, as indicated by its large R-value (.78). There are a few plausible explanations for this linear relationship. First, the elevated inflation rates of 2022, peaking at 9.1% in June (40-year high), may have increased elected officials concern about deficit spending. Public discontent over higher prices may have also pressured representatives to take a more hawkish approach on federal expenditures.

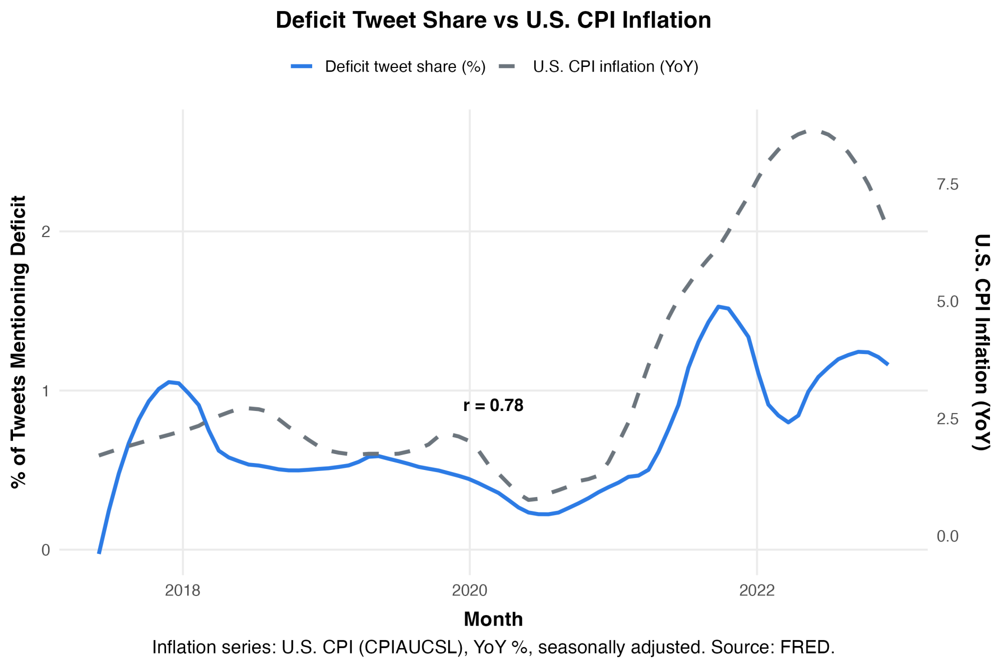


Figure 2: Deficit tweet share vs. U.S. CPI inflation.

Another explanation could be that along with higher inflation, the significant, and contentious, fiscal legislation being proposed in the summer of 2022 (e.g., IRA, CHIPS) may have drawn concerns from representatives, prompting more congressional tweets on the topic.

### 5.3 Member-Level Patterns

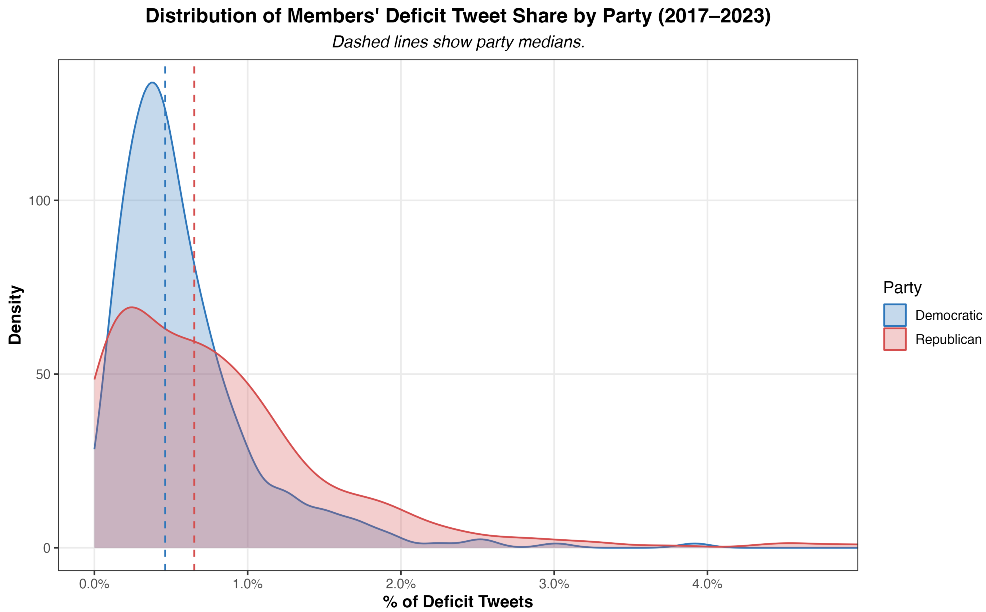


Figure 3: Share of deficit tweets by party (member-level distribution).

Figure 3 presents member-level distributions of deficit tweeting, calculated as the share of each member's tweets that mention the deficit over the entire period. As illustrated by the dashed lines (party medians), Republicans tweet about the deficit more frequently than Democrats, with average shares of 0.81% and 0.62%, respectively.

Democrats display a relatively compact distribution, with most representatives falling below 0.7%, indicating less intra-party variation in deficit tweeting. Republicans, however, show far greater variability within their party. The GOP's right tail distribution illustrates that, compared to their Democratic colleagues, a far greater proportion of members tweet about the deficit at much higher rates, with outliers reaching 4–8%.

Figure 4 breaks down these distributions further by institutional power status (majority vs. minority). The left and right panels juxtapose the distribution when a member's party controls the presidency (in-power) versus when it does not (out-of-power). For instance, Democrats would be considered in-power from January 2021 to January 2023 under President Joe Biden, while Republicans would be out-of-power during this same period.

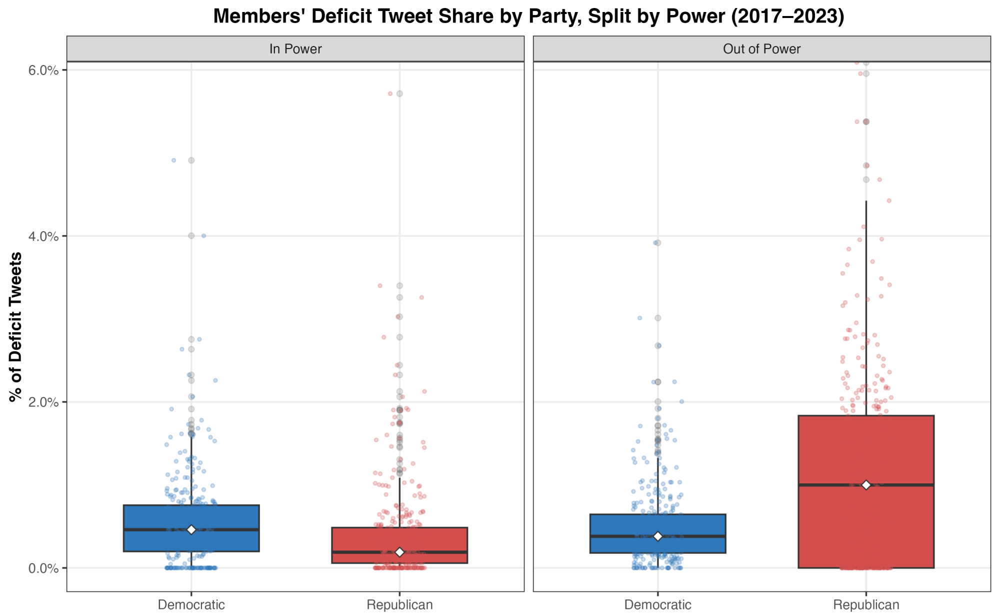


Figure 4: Distribution of deficit tweets by power (boxplot).

Figure 4 shows a clear asymmetry in Republican tweeting patterns between these periods. When moving between power contexts, the Republican median tweet share increases five-fold, from 0.2% to 1.0% and the interquartile range from 0.6% to 1.8%. Put simply, Republicans were far more likely to tweet about the deficit under President Biden than under President Trump. This stark distributional contrast illustrates a synchronized and cohesive shift in messaging among Republicans: emphasize the deficit more in periods of political weakness. Democrats, by contrast, remain relatively stable across contexts and in fact demonstrate a modest increase in spread and averages when in power.

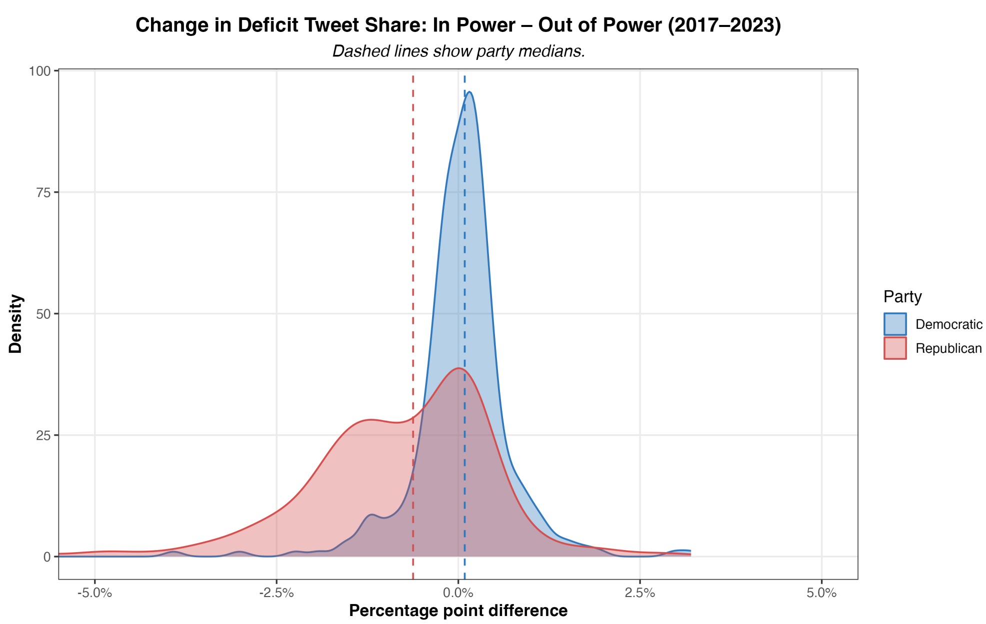


Figure 5: In–out shifts in deficit tweeting (histogram).

Figure 5 visualizes these shifts more directly by plotting the distribution of changes in deficit tweeting when moving from in-power to out-of-power contexts. The x-axis represents the difference in deficit tweet share between these two periods for each member (in minus out). Positive values indicate that a member tweeted more about the deficit when in power, while negative values indicate higher tweeting rates when out of power. The GOP's comprehensive party shift is evident, with the bulk of the party's distribution skewed negative.

Taken together, Figure 4 and Figure 5 suggest that Republicans are far more sensitive to power status when deciding how frequently to tweet about the deficit, while Democrats maintain a more consistent messaging strategy regardless of institutional control. These results also illustrate a broader point about the degree of strategic messaging coordination among parties: the systematic increase in Republican deficit tweeting supports the notion of uniform, top-down messaging within the party. Democrats, by contrast, show less cohesion in their messaging strategy, with members displaying a wider range of tweeting behaviors across power contexts.

## 5.4 Leadership Analysis

Figure 6 summarizes deficit-tweeting by House and Senate leaders across majority/minority contexts. Patterns differ by party and chamber, with the sharpest contrasts in the Senate: Senator Chuck Schumer tweeted about deficits far more as Majority Leader (4.0%) than as Minority Leader (1.0%), while Senator Mitch McConnell did the reverse (1.9% in the minority vs. 0.05% in the majority). In the House, Representative Pelosi tweeted about deficits more in the minority (1.1%) than as Speaker (0.5%).

**Deficit Tweeting Rates by Congressional Leaders (2017–2023)**

*Includes only leaders who served during this period*

| Leadership Role | Party | Total Tweets | Debt Tweets | % Deficit-Related |
|---|---|---|---|---|
| Chuck Schumer | | | | |
| Senate Majority Leader | Democratic | 4173 | 167 | 4.00 |
| Senate Minority Leader | Democratic | 8994 | 93 | 1.03 |
| Kevin McCarthy | | | | |
| House Minority Leader | Republican | 762 | 0 | 0.00 |
| Mitch McConnell | | | | |
| Senate Majority Leader | Republican | 5622 | 3 | 0.05 |
| Senate Minority Leader | Republican | 2483 | 47 | 1.89 |
| Nancy Pelosi | | | | |
| House Majority Leader | Democratic | 5617 | 28 | 0.50 |
| House Minority Leader | Democratic | 2705 | 30 | 1.11 |
| Paul Ryan | | | | |
| House Majority Leader | Republican | 3570 | 6 | 0.17 |

Figure 6: Deficit tweeting by congressional leaders (House and Senate; majority vs. minority roles).

Notably, Representative McCarthy "has relatively few tweets (762). Because leaders often switch or add handles when roles change, the data collection process may have failed to include relevant accounts. Thus, McCarthy's deficit tweet share should be interpreted cautiously.

Taken together, these patterns illustrate that deficit rhetoric among congressional leaders varies systematically with institutional role and party, mirroring—and in some cases, amplifying—the broader party-level trends documented above.

While preceding analysis covers broader party-level trends, exploring individual leaders' tweeting patterns provides us insight into how key political figures communicate about the deficit in varying institutional contexts. Considering the legislative and electoral significance of these figures, their messaging strategies are more influential and calculated than rank-and-file members. For instance, party leaders may use specific communication strategies around legislative periods to catalyze or bar the passage of key bills. Examining these trends can therefore help illustrate the degree of strategic communication employed by top legislators, and by extension — the party they represent.

Consider the differences in tweeting patterns among the two parties top legislators. First, for Democrats, the divergent tweeting patterns of Speaker Pelosi and Senator Schumer suggest a weak intra-party alignment in messaging. And, these patterns are consistent with the broader Democratic caucus, which shows no discernible difference in deficit tweeting between power contexts (as shown in Figure 4 and Figure 5).The Republican party, however, presents a more uniform party dynamic. Senator McConnell's increase in deficit tweeting in out-of-power contexts mirrors the broader Republican trend illustrated in Figure 5. Furthermore, McConnell's tweeting patterns confirm long-held conceptions about his communication and political style, which emphasizes unrelenting opposition to Democratic fiscal policies (MacGillis 2014; Hacker and Pierson 2016).

## 5.5 Legislative Context

Figure 7 reports the conditional share of tweets mentioning the deficit within $\pm$1-month windows around major bills, split by party (with a pooled/overall column for reference). At baseline, Republicans tweet about the deficit more often 0.8% than Democrats 0.6% – a gap that persists across most legislative windows. During the Inflation Reduction Act (2022) and CHIPS Act (2022) periods, Republicans' tweets accelerated, with deficit mentions at nearly 2.0%, compared to 1.3% among Democrats. However, the TCJA (2017)—a partisan GOP tax cut—is a noticeable and stark exception. During these three months, the Democrats' deficit share spikes to 2.0%, while the Republicans' share falls to just 0.2%, reflecting a partisan realignment around a highly divisive bill. In contrast, during the COVID relief period in 2020 – a non-partisan voting context – both parties converged at very low levels (0.3–0.4%).

## 5.6 Regression Visualization

The following regression models offer a structured approach to examining how institutional, fiscal, and partisan contexts affect the likelihood of congressional deficit tweeting. Each model includes interaction coefficients of the general form GOP $\times$ Conditioning Variable, indicating whether the baseline Republican–Democratic difference changes under specific conditions.

The baseline model, Figure 8, examines partisan differences in legislative periods without conditioning on institutional control. The non-significant results suggest that, when averaged across both Democratic and Republican presidencies, legislative windows themselves do not systematically impact the partisan gap in deficit tweeting.

Figure 9 controls for presidential party by adding an indicator for serving under the out-party president (GOP under a Democratic president; Democrats under a Republican president) and by interacting it with Legislative window, plus the three-way term GOP $\times$ Legislative window $\times$ Minority presidency. These terms let the GOP–Dem deficit-tweeting gap differ under Democratic vs. Republican presidents and specifically during legislative months.

| Deficit Share Within Each Window by Party (Tweet Level) | | | | |
|---|---|---|---|---|
| Conditional share of deficit tweets given the tweet is inside the window. | | | | |
| | | Tweets (N) | Deficit (N) | Deficit Rate (%) |
| All Tweets (Baseline) | Democratic | 2,138,163 | 13,199 | 0.6% |
| | Republican | 1,467,248 | 11,903 | 0.8% |
| Any Legislative Window | Democratic | 687,875 | 5,569 | 0.8% |
| | Republican | 486,104 | 4,558 | 0.9% |
| IRA / CHIPS Window | Democratic | 91,767 | 1,194 | 1.3% |
| | Republican | 74,031 | 1,421 | 1.9% |
| CARES Window | Democratic | 131,549 | 521 | 0.4% |
| | Republican | 82,587 | 231 | 0.3% |
| PPP–HCE Window | Democratic | 135,059 | 342 | 0.3% |
| | Republican | 88,440 | 250 | 0.3% |
| COVID Window (CARES or PPP–HCE) | Democratic | 171,383 | 627 | 0.4% |
| | Republican | 109,109 | 335 | 0.3% |
| TCJA Window | Democratic | 85,463 | 1,680 | 2.0% |
| | Republican | 62,372 | 139 | 0.2% |
| *Percentages are conditional on being inside each window; counts are tweet totals* ***within*** *the window.* | | | | |

Figure 7: Deficit share within legislative windows, by party.

| Legislative Windows and Major Bills | | |
|---|---|---|
| | Odds Ratio | 95% CI |
| GOP × Legislative window | 0.88 | [0.57, 1.36] |
| GOP × COVID window | 0.65 | [0.33, 1.28] |
| GOP × IRA/CHIPS window | 1.24 | [0.46, 3.32] |
| GOP × Bill partisanship (z) | 0.82 | [0.52, 1.29] |
| GOP × Fiscal magnitude (z) | 1.02 | [0.97, 1.07] |

Figure 8: Regression 1: Legislative windows.

Republicans' odds of tweeting about deficits are nearly double (OR = 1.87) when not holding the presidency (i.e., Joe Biden is President) compared to Democrats under the same condition. In contrast, GOP × Legislative window (OR= 0.38) highlights that under a Republican presidency, the partisan gap narrows during legislative months—the odds of the GOP tweeting are 62% lower than those of Democrats. In other words, Republicans are far less likely to discuss the deficit when their party holds the executive branch. This pattern aligns with Figure 1, which shows that Republican tweeting contracts while Democratic tweeting spikes as Congress passes the Tax Cuts and Jobs Act in late 2017. However, the three-way interaction coefficient GOP × Legislative window × Minority presidency reveals a reversal of this trend (OR= 4.24). Under a Democratic presidency (Joe Biden), Republicans' odds of tweeting are more than four times higher than those of Democrats in the same condition.

Figure 10 examines how the tweeting gap changes by who controls congressional and executive branches.

**Presidential Control and Legislative Windows**

| | Odds Ratio | 95% CI |
|---|---|---|
| GOP × Legislative window | 0.38*** | [0.26, 0.56] |
| GOP × Minority presidency | 1.87* | [1.07, 3.27] |
| GOP × Legislative window × Minority presidency | 4.24*** | [2.27, 7.92] |
| GOP × COVID window | 1.03 | [0.61, 1.74] |
| GOP × IRA/CHIPS window | 0.78 | [0.33, 1.85] |
| GOP × Bill partisanship (z) | 0.51*** | [0.41, 0.63] |
| GOP × Fiscal magnitude (z) | 0.99 | [0.98, 1.01] |

Figure 9: Regression 2: Majority context (presidency).

**Government Control and Legislative Windows**

| | Odds Ratio | 95% CI |
|---|---|---|
| GOP × Legislative window (GOP trifecta) | 0.31*** | [0.21, 0.46] |
| GOP × Legislative window × Dem trifecta | 7.00*** | [4.13, 11.85] |
| GOP × Legislative window × GOP Senate+Presidency | 0.85 | [0.47, 1.54] |
| GOP × COVID window | 1.10 | [0.62, 1.95] |
| GOP × IRA/CHIPS window | 0.79 | [0.33, 1.86] |
| GOP × Bill partisanship (z) | 0.50*** | [0.39, 0.64] |
| GOP × Fiscal magnitude (z) | 0.99 | [0.98, 1.01] |

Figure 10: Regression 3: Majority context — chamber and presidency combinations.

A GOP trifecta means Republicans held the presidency, House, and Senate simultaneously—a condition they held from 2017 to 2019 during the 115th congress. The model includes indicators for these control combinations: the baseline is Democrats outside legislative windows when their party controls no institution (minority in both chambers and not holding the presidency). This context would describe Democrats, between 2017 and 2019, in non-legislative periods (i.e, not during Tax Cuts and Jobs Act).

GOP x Legislative window (OR = 0.31) indicates that during legislative months, when holding a trifecta, Republicans' odds of tweeting are 69% lower than Democrats. The Tax Cuts and Jobs Act (2017) – a legislative window under a GOP trifecta – is an illustration of this condition. However, once we condition on a Democratic trifecta, this effect reverses. Relative to the baseline condition (a GOP trifecta), Republicans' odds of deficit tweeting are seven times higher than those of Democrats in the same context. The juxtaposition of these two coefficients (OR = 0.31) versus (OR = 7.00) illustrates that legislative timing by itself isn't a determinate of tweeting gaps—but who governs is. Under Democratic control (i.e., 2021-2023), Republicans accelerate deficit rhetoric during legislative months, while under the GOP trifecta (i.e., 2017-2019), they significantly reduce deficit tweeting frequencies.

### 5.6.1 Bill Partisanship and Fiscal Magnitude

Across all three specifications, the coefficients for Fiscal magnitude (z) are statistically insignificant (~1.0). These coefficients, based on a normalized measure of a given bill's 10-year CBO estimate, suggest that the fiscal consequences of policy do not systematically impact partisan tweeting gaps.

Bill partisanship (z), however, shows significant contractions in the GOP-Dem tweeting gap once institutional control is included (OR ~ 0.50). In other words, for each one-SD increase in a bill's partisanship score, the GOP-Dem gap contracts by roughly half. Notably, because these variables are continuous (i.e., z-scores), the coefficients represent slopes rather than a baseline-specific effect. For example, in the context of bill partisanship, if Republicans are twice as likely as Democrats to tweet at $z = 0$ ($OR = 2.0$), that gap narrows to parity at $z = 1$ ($OR = 1.0$).